Communication Depth in Multivalued Broadcast

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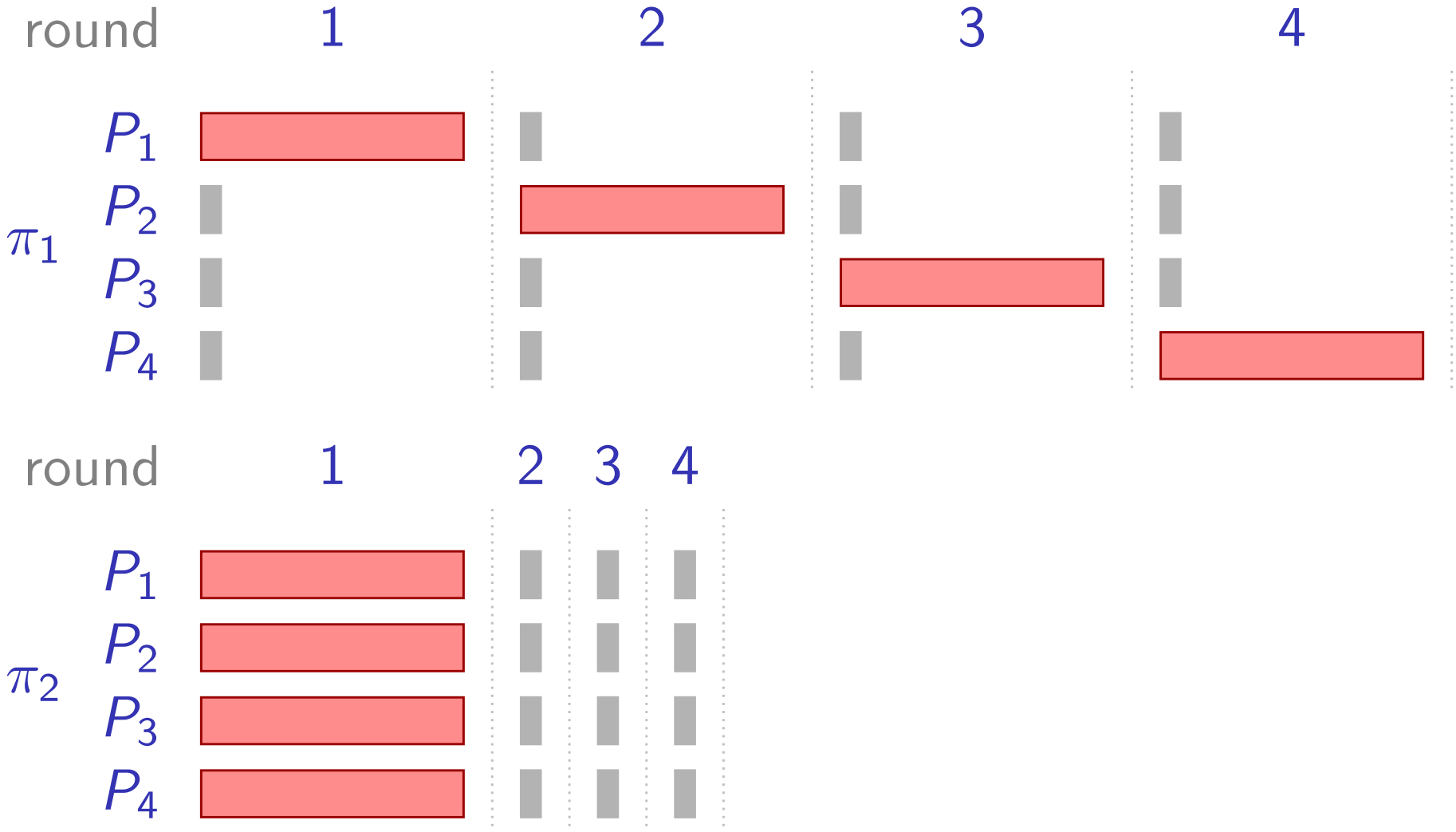
ETH Zürich

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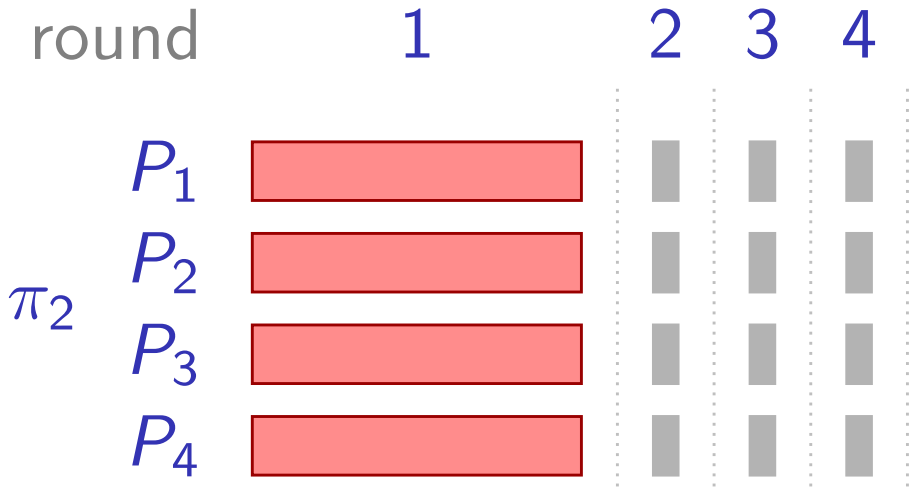
Efficiency of Synchronous Protocols

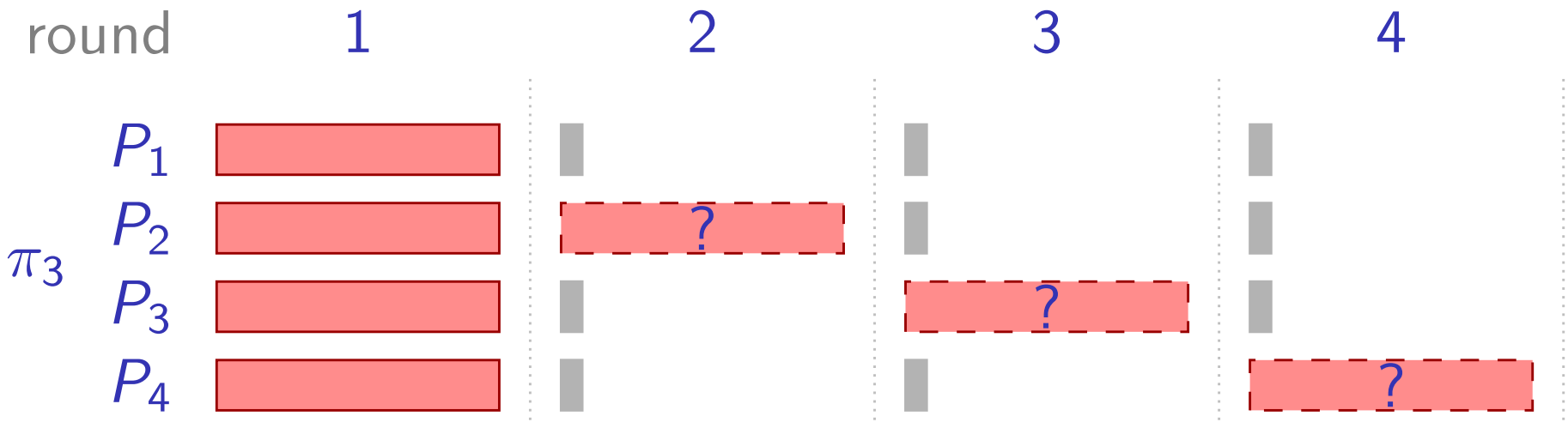
- Round Complexity
- Message Complexity
- Communication Complexity
- Balanced Communication Complexity
- ...

Equally Efficient?



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Communication in Synchronous Protocols

- Communication may be parallelized within rounds
- The duration of a round depends on amount of data transmitted
- Parties must agree on this duration in advance to remain synchronized

Communication Depth

Definition: Communication Depth of a protocol Π

- Write $\ell_{ij}^r(\vec{x}, \vec{\rho}, \mathcal{A})$ for the amount of data that Π requires P_i to send to P_j in round r when executed with inputs $\vec{x}$, randomness $\vec{\rho}$ and adversary $\mathcal{A}$.
- Define

$$D := \sum_r \max_{\vec{x},\vec{\rho},\mathcal{A}} \max_{i,j} \ell_{ij}^r(\vec{x}, \vec{\rho}, \mathcal{A})$$

Broadcast

Broadcast Definition

- Sender P_S chooses message to send to all parties
- Even if P_S is corrupt, all honest parties receive the same message

Communication

- Comm. Complexity Lower Bound: $\Omega(\ell n)$
- Comm. Depth Lower Bound: $\Omega(\frac{\ell}{n-t})$

Overview: Comm. Depth for $\ell \gg n$

	Threshold	Security	D
Chen21 [Che21]	$t < n/3$	perfect	$\mathcal{O}(\ell/n)$
Nay$^+$20 [Nay+20]	$t < n/2$	cryptogr.	$\mathcal{O}(\ell/n)^*$
FH06 [FH06]	$t < n/2$	inf.-theor.	$\mathcal{O}(\ell/n)^*$
CO18 [CO18]	$t < n$	inf.-theor.	$\mathcal{O}(\ell)$
GP21 [GP21]	$t < n$	cryptogr.	$\mathcal{O}(\ell)$
Nay$^+$20 [Nay+20]	$t < (1-\epsilon)n$	cryptogr.	$\mathcal{O}(\ell)$
This Paper	$t < (1-\epsilon)n$	cryptogr.	$\mathcal{O}(\ell/n)$
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$$B_k = \{P \mid d(P_s, P) \leq k\}$$

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(D) Apply the propagation protocol up to $\frac{2n}{n-t}$ times.

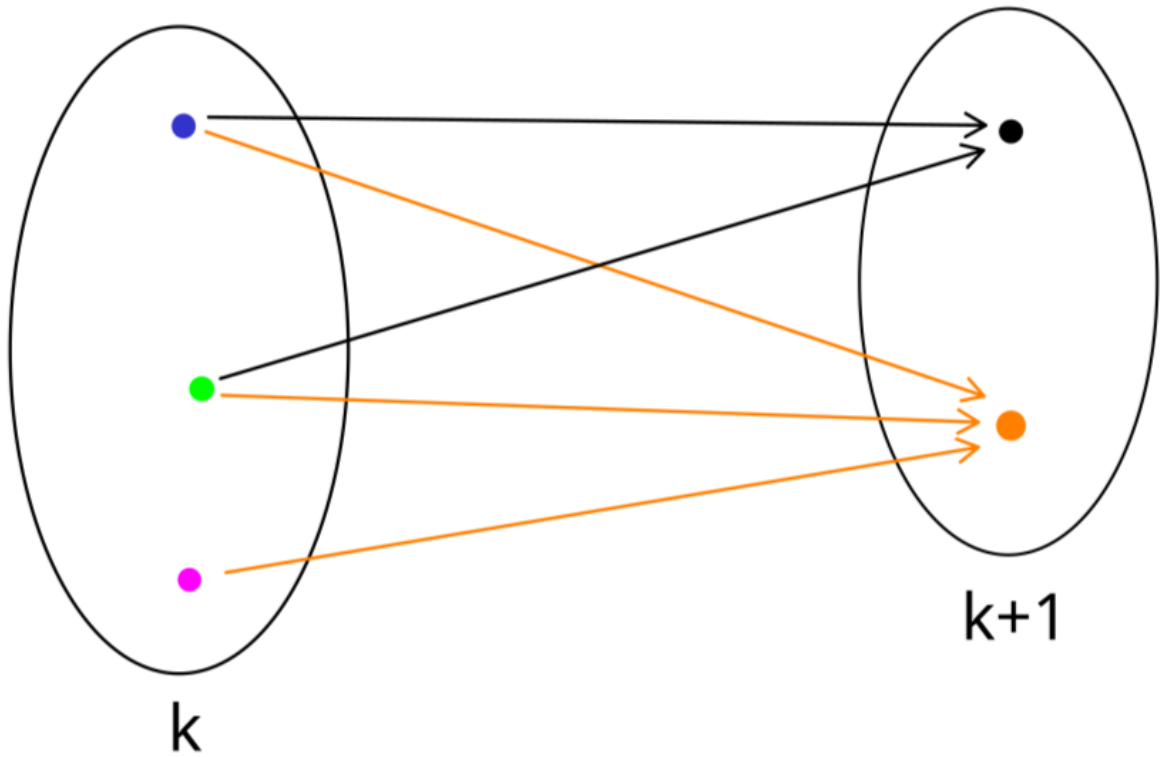
Propagation Protocol

Encode m into shares $m_1, \ldots m_n$ of size $\frac{\ell}{n-t}$ (Reed-Solomon code)

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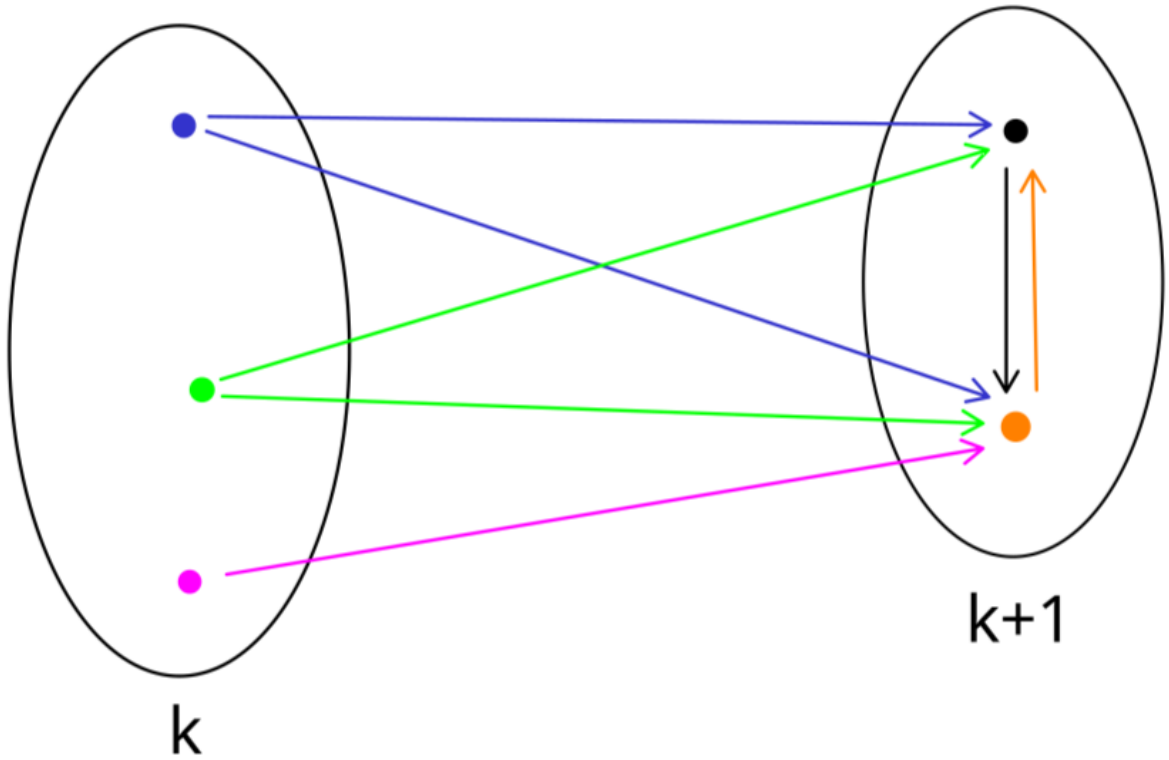
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Fact: Every P_j which started out at distance $k+1$ has either

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Communication Depth: $\mathcal{O}(\frac{\ell}{n-t} + \text{poly}(n))$

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Propagation $\Rightarrow$ Broadcast

- split message into $q := \frac{2n}{n-t}$ blocks
- staggered delay: start i-th block after i iterations
- parallelization without increased Communication Depth: only one block is propagated from distance k to $k+1$

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Thm:
Assuming PKI for signatures, and a collision resistant hash function; for any constant $\epsilon > 0$, broadcast of an ℓ-bit message can be achieved with communication depth $\mathcal{O}(\ell/n + \kappa n^3)$ and computational security up to $t < (1-\epsilon)n$ corruptions.

References I

[BH06] Zuzana Beerliova-Trubiniova and Martin Hirt. “Efficient Multi-Party Computation with Dispute Control”. *Theory of Cryptography Conference — TCC 2006*. Ed. by Shai Halevi and Tal Rabin. Vol. 3876. Lecture Notes in Computer Science. Springer-Verlag, Mar. 2006, pp. 305–328.

References II

[Che21] Jinyuan Chen. "Optimal Error-Free Multi-Valued Byzantine Agreement". *35th International Symposium on Distributed Computing (DISC 2021)*. Ed. by Seth Gilbert. Vol. 209. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021, 17:1–17:19. ISBN: 978-3-95977-210-5. DOI: 10.4230/LIPIcs.DISC.2021.17. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2021.17.

References III

[CO18] Wutichai Chongchitmate and Rafail Ostrovsky. "Information-Theoretic Broadcast with Dishonest Majority for Long Messages". *Theory of Cryptography: 16th International Conference, TCC 2018, Panaji, India, November 11–14, 2018, Proceedings, Part I*. Panaji, India: Springer-Verlag, 2018, pp. 370–388. ISBN: 978-3-030-03806-9. DOI: 10.1007/978-3-030-03807-6_14. URL: https://doi.org/10.1007/978-3-030-03807-6_14.

[FH06] Matthias Fitzi and Martin Hirt. "Optimally Efficient Multi-Valued Byzantine Agreement". *Proc. 25th ACM Symposium on Principles of Distributed Computing — PODC 2006*. Ed. by Dahlia Malkhi. ACM, July 2006.

References IV

[GP21] Chaya Ganesh and Arpita Patra. "Optimal extension protocols for Byzantine broadcast and agreement". *Distributed Computing* 34 (Feb. 2021). DOI: 10.1007/s00446-020-00384-1.

[Nay+20] Kartik Nayak et al. "Improved extension protocols for Byzantine broadcast and agreement". English (US). *34th International Symposium on Distributed Computing, DISC 2020*. Ed. by Hagit Attiya. Leibniz International Proceedings in Informatics, LIPIcs. Schloss Dagstuhl-Leibniz-Zentrum fur Informatik GmbH, Dagstuhl Publishing, Oct. 2020. DOI: 10.4230/LIPIcs.DISC.2020.28.

References V

[Wan+20] Jun Wan et al. “Expected Constant Round Byzantine Broadcast Under Dishonest Majority”. *TCC 2020, Part I*. Ed. by Rafael Pass and Krzysztof Pietrzak. Vol. 12550. LNCS. Springer, Cham, Nov. 2020, pp. 381–411. DOI: 10.1007/978-3-030-64375-1_14.